

ARANYA SAHA

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EDUCATION

University of Maryland, College Park

Aug 2026 – Present

PhD in Electrical and Computer Engineering

Research Advisor: *Prof. Sanghamitra Dutta*

Bangladesh University of Engineering and Technology

Feb 2020 – Mar 2025

BSc in Electrical and Electronic Engineering

Research Advisor: *Prof. Mohammad Ariful Haque*

WORK EXPERIENCE

University of Maryland, College Park

Aug 2026 – Present

Graduate Research Assistant

MD, USA

Advanced Chemical Industries PLC

Apr 2025 – Aug 2026

Machine Learning Engineer

Dhaka, Bangladesh

Bangladesh University of Engineering and Technology

Nov 2025 – Apr 2026

Part-time Lecturer, Electrical and Electronic Engineering

Dhaka, Bangladesh

Institute of Robotics and Automation, BUET

Jun 2025

Instructor, Robotics Bootcamp 2025

Dhaka, Bangladesh

RESEARCH INTERESTS

Efficient AI, Mechanistic Interpretability, Reinforcement Learning, Large Language Models, AI for Science.

PUBLICATIONS

Preprints

1. Ismam Nur Swapnil, **Aranya Saha**, Tanvir Ahmed Khan, Mohammad Ariful Haque, Ser-Nam Lim, “Gradient Extrapolation-Based Policy Optimization.” – [Link](#)
2. **Aranya Saha***, Tanvir Ahmed Khan*, Ismam Nur Swapnil*, Mohammad Ariful Haque, “CLARIFY: A Specialist-Generalist Framework for Accurate and Lightweight Dermatological Visual Question Answering.” – [Link](#)
3. Ismam Nur Swapnil*, **Aranya Saha***, Tanvir Ahmed Khan*, Mohammad Ariful Haque, “GRPO++: Enhancing Dermatological Reasoning under Low Resource Settings.” – [Link](#)
4. Tanvir Ahmed Khan, **Aranya Saha**, Ismam Nur Swapnil, Mohammad Ariful Haque, “Compression Strategies for Efficient Multimodal LLMs in Medical Contexts.” – [Link](#)

Conferences

1. Shadman Sobhan, **Aranya Saha**, Tanvir Ahmed Khan, Abduz Zami, “Skin Cancer Classification Using Pre-trained CNNs: A Transfer Learning Approach Addressing Imbalanced Data Challenges,” International Conference on Next-Generation Computing, IoT and Machine Learning (2025). – [Link](#)
2. Shadman Sobhan, Abduz Zami, Mohiuddin Ahmed, Tanvir Mahtab Zihan, Tanvir Ahmed Khan, **Aranya Saha**, “A Multi-Stage Deep Learning Approach to Tuberculosis Detection with Explainable Insights,” International Conference on Next-Generation Computing, IoT and Machine Learning (2025). – [Link](#)

SELECTED PROJECTS

1. **Efficient Frame Selection for Long Egocentric Video Understanding.** Developed an efficient frame selection approach for long-form egocentric video understanding. – GitHub
2. **EchoLens: Multimodal Conversational AI Engine.** Developed a multimodal conversational AI system integrating vision and language models for interactive understanding and response generation. – GitHub
3. **Simple MedQA-GPT: GPT Tailored for Medical Q&A.** Developed a GPT-based system for answering medical questions with domain-specific prompting and conversational interaction. – GitHub
4. **IoT-Based Patient Health Monitoring System.** Developed an IoT-enabled system for real-time monitoring of patient health parameters. – GitHub Report
5. **MATLAB-Based Fingerprint Recognition System.** Developed a fingerprint recognition system using MATLAB for biometric identification and matching. – GitHub Report

RELEVANT COURSEWORK

Graduate	Random Processes, Advanced Digital Signal Processing
Undergraduate	Artificial Intelligence and Machine Learning, Digital Signal Processing, Random Signals and Processes, Probability and Statistics, Signals and Systems, etc.

TECHNICAL SKILLS

Programming	Python, C/C++, MATLAB
ML/DL	PyTorch, TensorFlow, CNNs, VLMs, LLMs
Data/Tools	NumPy, Pandas, Git, Docker, FastAPI, LaTeX
Hardware	Microcontrollers, IoT Devices, Sensors

LEADERSHIP EXPERIENCE

Association for Computing Machinery (ACM) – SIGCOMM	Apr 2024 – Feb 2025
<i>Student Executive</i>	<i>Remote</i>
Served as the first Student Executive of ACM SIGCOMM, working alongside Prof. Matthew Caesar (UIUC), Chair of ACM SIGCOMM; developed its official website under his mentorship and co-established a paper reading group.	

HONORS AND AWARDS

- Dean's Fellowship (ECE) – University of Maryland, College Park (Fall 2026).
- 1st Runner-Up, Poster Competition (AI) – BEAR Summit 2025, Poster Link, Certificate Link
- University Merit Scholarship – BUET, awarded for academic performance (4 semesters).
- Dean's List Award – BUET, awarded for academic performance (Years 1–2).
- Ranked 29th among 10,000+ candidates in the BUET Undergraduate Admission Test (2019).
- Ranked 31st (Male) in the Dhaka Board HSC Examination; recipient of the Talent Pool Scholarship.
- Perfect Attendance Certificate – Notre Dame College.

REFERENCES

Prof. Sanghamitra Dutta Electrical and Computer Engineering University of Maryland, College Park Email: sanghamd@umd.edu	Prof. Mohammad Ariful Haque Electrical and Electronic Engineering Bangladesh University of Engineering and Technology Email: arifulhoque@eee.buet.ac.bd
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